

Pair Sum Problem:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter array size: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter array elements:");
```

```
        for(int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        System.out.print("Enter target value: ");
```

```
        int target = sc.nextInt();
```

```
        boolean found = false;
```

```
        for(int i = 0; i < arr.length; i++) {
```

```
            for(int j = i + 1; j < arr.length; j++) {
```

```

        if(arr[i] + arr[j] == target) {
            System.out.println(arr[i] + " + " + arr[j] + " = " + target);
            found = true;
        }
    }
}

if(!found) {
    System.out.println("No pair found");
}

sc.close();
}
}

```

Triplet Sum Problem:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
    }
}

```

```
System.out.println("Enter array elements:");  
for(int i = 0; i < n; i++) {  
    arr[i] = sc.nextInt();  
}
```

```
System.out.print("Enter target value: ");  
int target = sc.nextInt();
```

```
boolean found = false;
```

```
for(int i = 0; i < arr.length; i++) {
```

```
    for(int j = i + 1; j < arr.length; j++) {
```

```
        if(arr[i] + arr[j] == target) {
```

```
            System.out.println(arr[i] + " + " + arr[j] + " = " + target);
```

```
            found = true;
```

```
        }
```

```
    }
```

```
}
```

```
if(!found) {
```

```
    System.out.println("No pair found");
```

```
}
```

```
sc.close();
```

```
}  
}
```

Maximun Subarray Sum:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter array size: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter array elements:");
```

```
        for(int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        int currentSum = arr[0];
```

```
        int maxSum = arr[0];
```

```
        for(int i = 1; i < n; i++) {
```

```

        currentSum = Math.max(arr[i], currentSum + arr[i]);

        maxSum = Math.max(maxSum, currentSum);
    }

    System.out.println("Maximum Subarray Sum = " + maxSum);

    sc.close();
}
}

```

Move Zero to End:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();

```

```
}
```

```
int index = 0;
```

```
// Move non-zero elements to front
```

```
for(int i = 0; i < n; i++) {
```

```
    if(arr[i] != 0) {
```

```
        arr[index] = arr[i];
```

```
        index++;
```

```
    }
```

```
}
```

```
// Fill remaining positions with zero
```

```
while(index < n) {
```

```
    arr[index] = 0;
```

```
    index++;
```

```
}
```

```
System.out.println("Array after moving zeros to end:");
```

```
for(int i = 0; i < n; i++) {
```

```
    System.out.print(arr[i] + " ");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

Merge Two Sorted Arrays:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n1 = sc.nextInt();
```

```
        int[] arr1 = new int[n1];
```

```
        for(int i = 0; i < n1; i++) {
```

```
            arr1[i] = sc.nextInt();
```

```
        }
```

```
        int n2 = sc.nextInt();
```

```
        int[] arr2 = new int[n2];
```

```
        for(int i = 0; i < n2; i++) {
```

```
            arr2[i] = sc.nextInt();
```

```
        }
```

```
        int[] result = new int[n1 + n2];
```

```
        int i = 0, j = 0, k = 0;
```

```

while(i < n1 && j < n2) {

    if(arr1[i] <= arr2[j]) {
        result[k++] = arr1[i++];
    } else {
        result[k++] = arr2[j++];
    }
}

while(i < n1) {
    result[k++] = arr1[i++];
}

while(j < n2) {
    result[k++] = arr2[j++];
}

for(int x : result) {
    System.out.print(x + " ");
}
}
}

```

Equilibrium Index:

```
import java.util.Scanner;
```

```
public class Main {
```



```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");
    int n = sc.nextInt();

    int[] arr = new int[n];

    System.out.println("Enter array elements:");
    for(int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();
    }

    int totalSum = 0;

    for(int i = 0; i < n; i++) {
        totalSum += arr[i];
    }

    int leftSum = 0;
    boolean found = false;

    for(int i = 0; i < n; i++) {

        totalSum = totalSum - arr[i];
```

```

        if(leftSum == totalSum) {
            System.out.println("Equilibrium Index = " + i);
            found = true;
            break;
        }

        leftSum += arr[i];
    }

    if(!found) {
        System.out.println("No Equilibrium Index Found");
    }

    sc.close();
}
}

```

Problem Statement: Reverse a String

Given a string **S**, write a program to reverse the string and print the reversed string.

Input Format

- A single line containing the string **S**.

Output Format

- Print the reversed string.

Constraints

- $1 \leq \text{Length of } S \leq 1000$

Sample Input 2

Java

Sample Output 2

avaJ

Explanation

The reverse of "Java" is "avaJ".

Sample Input 3

Programming

Sample Output 3

gnimmargorP

Java Program

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String reverse = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reverse += str.charAt(i);
        }

        System.out.println(reverse);
    }
}
```

Problem Statement: Palindrome String

Given a string **S**, check whether it is a palindrome or not.

A string is called a **palindrome** if it reads the same forwards and backwards.

If the string is a palindrome, print "**Palindrome**". Otherwise, print "**Not Palindrome**".

Input Format

- A single line containing the string **S**.

Output Format

- Print "**Palindrome**" if the string is a palindrome.
- Otherwise, print "**Not Palindrome**".

Constraints

- $1 \leq \text{Length of } S \leq 1000$
- String contains only alphabets.

Sample Input 1

madam

Sample Output 1

Palindrome

Explanation

Original String = "madam"

Reversed String = "madam"

Both are equal, so it is a palindrome.

Sample Input 2

hello

Sample Output 2

Not Palindrome

Explanation

Original String = "hello"

Reversed String = "olleh"

Both are not equal, so it is not a palindrome.

Sample Input 3

racecar

Sample Output 3

Palindrome

Java Program

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String reverse = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reverse += str.charAt(i);
        }

        if (str.equals(reverse)) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }
    }
}
```

Problem Statement: Count Vowels and Consonants

Given a string **S**, write a program to count the number of vowels and consonants present in the string.

Vowels

The vowels are:

a, e, i, o, u

A, E, I, O, U

All other alphabets are considered consonants.

Input Format

- A single line containing the string **S**.

Output Format

Print the count of vowels and consonants in the following format:

Vowels: X

Consonants: Y

Where:

- X = Number of vowels
- Y = Number of consonants

Constraints

- $1 \leq \text{Length of S} \leq 1000$
- String contains only English alphabets.

Sample Input 1

hello

Sample Output 1

Vowels: 2

Consonants: 3

Explanation

String = "hello"

Vowels = e, o → 2

Consonants = h, l, l → 3

Sample Input 2

Programming

Sample Output 2

Vowels: 3

Consonants: 8

Explanation

Vowels = o, a, i → 3

Consonants = P, r, g, r, m, m, n, g → 8

Sample Input 3

AEIOU

Sample Output 3

Vowels: 5

Consonants: 0

Java Program

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String str = sc.nextLine();
```

```
        int vowels = 0;
```

```
        int consonants = 0;
```

```
for (int i = 0; i < str.length(); i++) {  
  
    char ch = Character.toLowerCase(str.charAt(i));  
  
    if (ch == 'a' || ch == 'e' || ch == 'i' ||  
        ch == 'o' || ch == 'u') {  
        vowels++;  
    } else {  
        consonants++;  
    }  
}  
  
System.out.println("Vowels: " + vowels);  
System.out.println("Consonants: " + consonants);  
}  
}
```